



GENOSPHERE

Clinical Genome Center — Variant Annotation & Interpretation Pipeline

COMPARATIVE VARIANT INTERPRETATION REPORT

Four Rare Monogenic Disease Variants — Synthetic Validation Dataset

Report Type	Comparative Pathogenic Variant Interpretation	Reference Build	GRCh38
Diseases Assessed	FOP, Apert Syndrome, Hereditary Hemochromatosis, AATD	Variant Class	Single Nucleotide Variants (SNVs) — Missense
Annotation Tools	Ensembl VEP 116.0, SnpEff/SnpSift, ANNOVAR	Classification Framework	ACMG/AMP Guidelines (InterVar)

Comparative Variant Table

Parameter	Fibrodysplasia Ossificans Progressiva (FOP)	Apert Syndrome	Hereditary Hemochromatosis	Alpha-1 Antitrypsin Deficiency (AATD)
VARIANT SUMMARY				
Gene	ACVR1	FGFR2	HFE	SERPINA1
Disease	Fibrodysplasia Ossificans Progressiva	Apert Syndrome	Hereditary Hemochromatosis	Alpha-1 Antitrypsin Deficiency
Chromosome	chr2	chr10	chr6	chr14
Position (GRCh38)	157,774,114	As reported in VCF	26,091,179	As reported in VCF
dbSNP ID	rs121912678	As reported in VCF	rs1800562	rs28929474
HGVS (DNA)	c.617G>A	c.755C>G	c.845G>A	c.1096G>A
HGVS (Protein)	p.Arg206His	p.Ser252Trp	p.Cys282Tyr	p.Glu366Lys
Variant Type	Missense Variant	Missense Variant	Missense Variant	Missense Variant
Consequence	Moderate Impact	Moderate Impact	Moderate Impact	Moderate Impact
QUAL	100	100	100	100
FILTER	PASS	PASS	PASS	PASS
Genotype	1/1 (Homozygous)	1/1 (Homozygous)	1/1 (Homozygous)	1/1 (Homozygous)
CLINICAL INTERPRETATION				
ClinVar	Pathogenic	Pathogenic	Pathogenic	Pathogenic
ACMG (InterVar)	Pathogenic	Pathogenic	Pathogenic	Likely Pathogenic
Disease Association	Strongly associated with FOP	Strongly associated with Apert Syndrome	Well-established cause of Hereditary Hemochromatosis (HFE1)	Well-established cause of AATD (Pi*Z allele)
FUNCTIONAL PREDICTION				

Parameter	Fibrodysplasia Ossificans Progressiva (FOP)	Apert Syndrome	Hereditary Hemochromatosis	Alpha-1 Antitrypsin Deficiency (AATD)
SIFT	Deleterious (0.03)	Deleterious	Deleterious	Deleterious
PolyPhen-2	Probably Damaging (0.999)	Probably Damaging	Probably Damaging	Probably Damaging
REVEL	Reported in VCF	High damaging score	Damaging (multiple in-silico concordance)	0.686
AlphaMissense	Reported in VCF	Pathogenic	Likely Pathogenic	0.5464 (Pathogenic)
CADD PHRED	Reported in VCF	High (>20)	High (>20)	25.7
SpliceAI	No significant splice effect	No significant splice effect	No significant splice effect	No predicted splice effect (0.00)
POPULATION DATA				
gnomAD Allele Frequency	Very rare / absent	Very rare	0.0338 (gnomAD v2.1, overall); up to 5.6% in European non-Finnish	0.0121565
OVERALL INTERPRETATION				
Final Classification	Pathogenic	Pathogenic	Pathogenic	Likely Pathogenic / Pathogenic evidence
Clinical Significance	Disease-causing ACVR1 mutation consistent with Fibrodysplasia Ossificans Progressiva.	Variant is strongly associated with Apert Syndrome and supported by multiple prediction tools.	The HFE p.Cys282Tyr variant is the most common pathogenic mutation causing hereditary hemochromatosis; homozygosity confers increased risk of iron overload affecting the liver, pancreas, heart, and joints.	The SERPINA1 p.Glu366Lys (Pi*Z) variant is a well-established disease-causing mutation associated with Alpha-1 Antitrypsin Deficiency; functional prediction tools consistently support a damaging effect.

Test Methodology

- Sequence data (synthetic VCF, GRCh38-aligned) annotated using Ensembl VEP 116.0 with cache-based transcript, consequence, and REVEL plugin annotation.
- Cross-validation performed using SnpEff/SnpSift and ANNOVAR for concordance across gene, HGVS nomenclature, and consequence prediction.
- Pathogenicity classification derived from ClinVar accessions and ACMG/AMP criteria applied via InterVar.
- Functional impact estimated using SIFT, PolyPhen-2, REVEL, AlphaMissense, CADD PHRED, and SpliceAI scores.
- Population frequency cross-checked against gnomAD to assess rarity consistent with autosomal dominant/recessive disease models.
- All four variants represent point mutations (SNVs); structural/CNV annotation branches (e.g., MAVERICK, Horizon) were not applicable and were disabled.

Recommendations

- Clinical correlation with patient phenotype is recommended prior to any diagnostic use.
- Genetic counseling is advised to discuss inheritance pattern, penetrance, and family screening implications for each disease.
- Confirmatory orthogonal testing (e.g., Sanger sequencing) is recommended before clinical reporting.
- This report is generated from a synthetic/validation VCF dataset for pipeline benchmarking and is not derived from an actual patient specimen.

Test Limitations

This analysis is limited to single nucleotide variants and small insertions/deletions at the four target loci; it does not assess copy number variants, structural rearrangements, or repeat expansions. Population frequency and in-silico prediction values reflect the most recent public database releases available at the time of pipeline configuration and may be updated as databases evolve. Classifications should be re-evaluated periodically in line with updated ACMG/AMP evidence.

Regulatory Disclosures

This report was generated by the Genosphere variant annotation pipeline (VEP 116.0 / SnpEff / ANNOVAR / InterVar stack) for research and educational purposes. It has not been reviewed for clinical diagnostic use and is not a substitute for an accredited clinical laboratory report.